Supply in a Competitive Market

Chapters 8

> **Perfect Competition:** A market (or industry) in which many firms produce an identical product and there are no barriers to entry.

1 Market Structures and Perfect Competition in the Short-Run

> **Market Structure:** The competitive environment in which firms operate.

We categorize a market or industry using three primary characteristics:

- Number of firms. Generally, the more companies in the market, the more competitive it is.
- Whether the consumer cares which company made the good. In general, the more indistinguishable or identical the products are, the more competitive the market is.
- Barriers to entry. If new firms can enter a market easily, the market is more competitive.

1.1 Perfect Competition

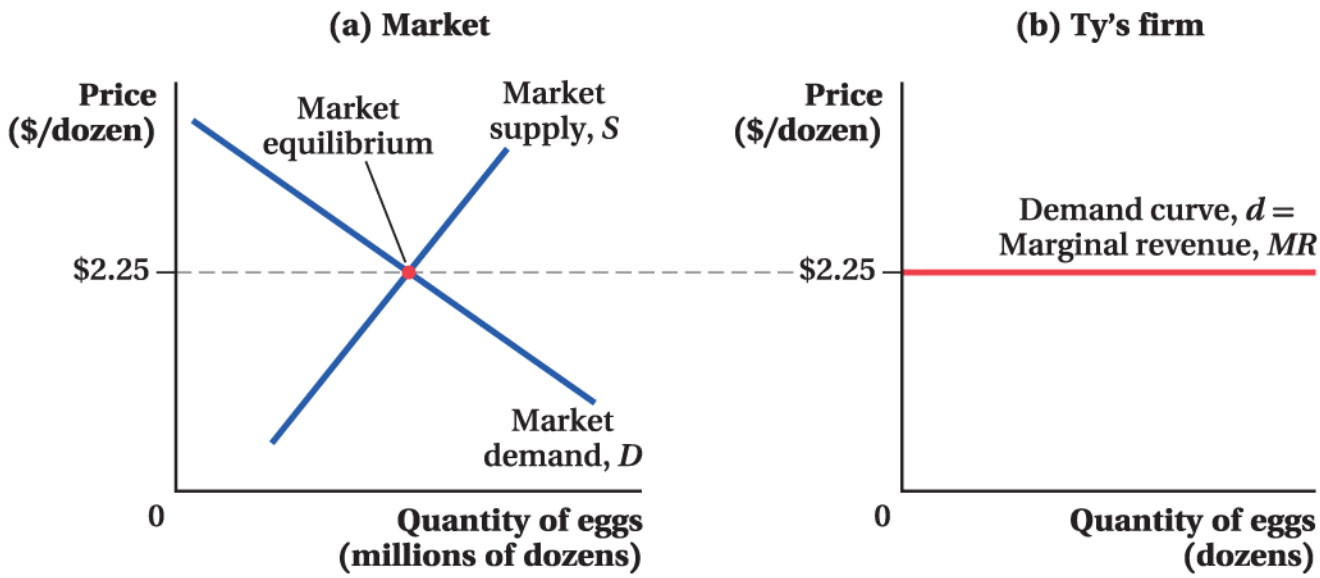
- **Number of Firms:** There needs to be a large number of firms so that no one firm has any impact on the market equilibrium price by itself. Thus, any one firm can change its behavior without changing the overall market equilibrium.
- **Types of Products Sold:** All firms produce an identical product. By identical, we don't just mean that all the firms make televisions or all the firms make smoothies. We mean the consumers view the output of the different producers as perfect substitutes and do not care who made it. This assumption is more accurate for nails or gasoline or bananas, but less accurate for smartphones or automobiles.
- **Barriers to Entry:** There are no barriers to entry. If someone decides she wants to start selling nails tomorrow, nothing prevents her from doing it.

An example of a perfectly competitive market is a commodity crop like soybeans.

1.2 Why Perfectly Competitive Markets Are Important to Study, Even if Rare in Real Life

- There are many markets that are almost perfectly competitive, and the perfect competition model provides a fairly good idea about how these markets work, too.
- Perfect competition is a benchmark against which economists measure the efficiency of other market structures.

1.3 The Demand Curve as Seen by a Price Taker



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- If the price taker charged a price higher than the market price they would sell zero units.
- Price taker would not charge a price lower than the market price, because they would make more money if they charged the market price.

POLL EVERYWHERE: The perfectly competitive firm's demand curve is

2 Profit Maximization in a Perfectly Competitive Market

> **Profit (Π):** The difference between a firm's revenue and its total cost.

2.1 Total Revenue, Total Cost, and Profit Maximization

As the definition above told us:

$$\Pi = TR - TC$$

Next, we know that marginal cost is equal to:

$$MC = \Delta TC / \Delta Q$$

> **Marginal Revenue:** The additional revenue from selling one additional unit of output.

$$MR_{PC} = \frac{\Delta TR}{\Delta Q} = \frac{\Delta(P \times Q)}{\Delta Q} = P\frac{\Delta Q}{\Delta Q} = P$$

In a perfectly competitive market, marginal revenue equals the market price; that is, $MR = p^*$.

Consider total revenue (proportional to its output because price is constant), marginal revenue, and marginal cost.

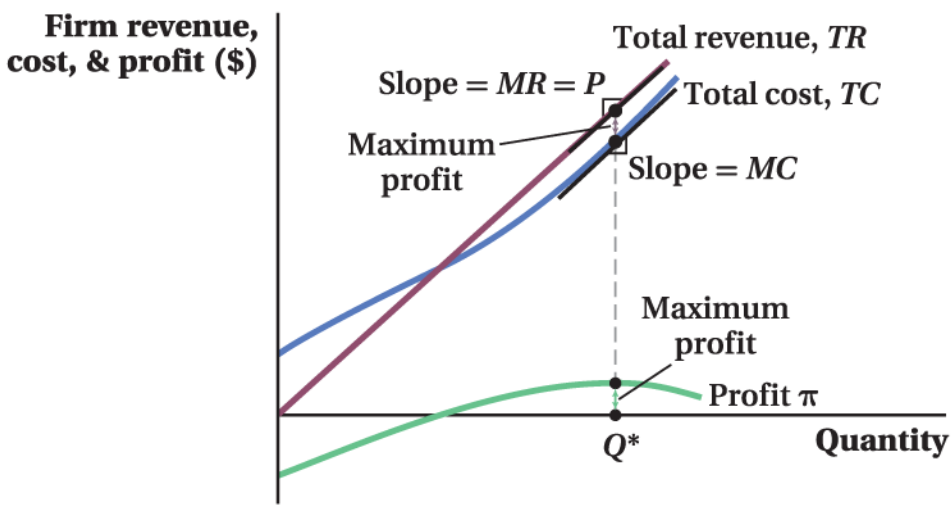

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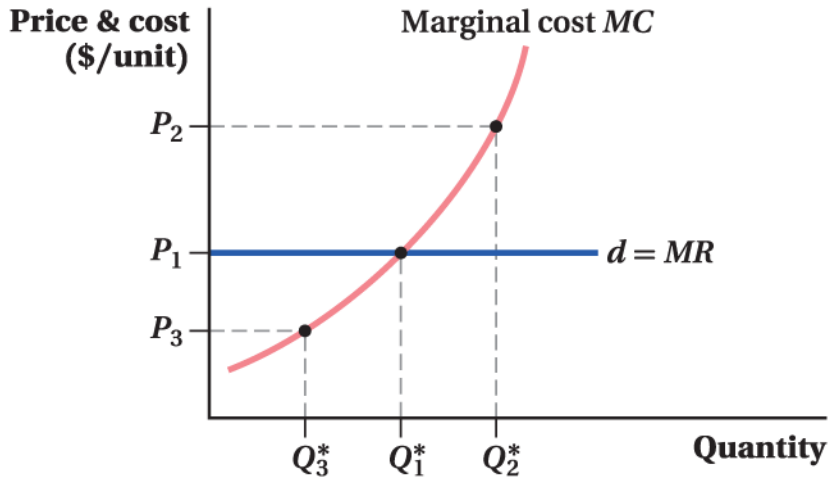
Figure 1: Because a perfectly competitive firm faces a constant market price, its total revenue curve is a straight line from the origin with a slope equal to marginal revenue, or the price. At the quantity Q^*, slope of the total revenue curve (price) equals the slope of the total cost curve (marginal cost), and the firm is maximizing profit.

POLL EVERYWHERE: A firm should _____ output whenever MR exceeds MC because _____.

2.2 How a Perfectly Competitive Firm Maximizes Profit

- $MR = P > MC$: Increase production, profits increase.
- $MR = P < MC$: Decrease production, profits increase.
- $MR = P = MC$: Profit Maximized



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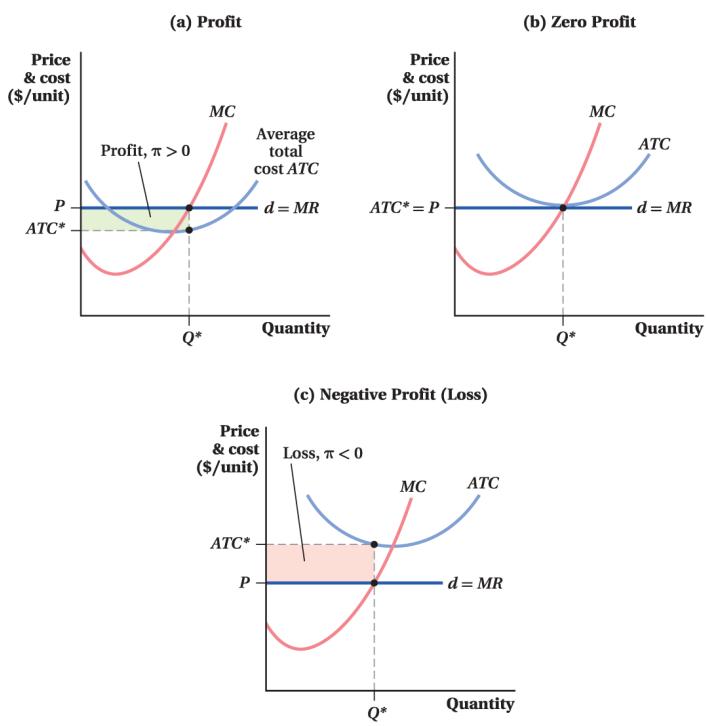
2.3 Measuring a Firm's Profit

$$\begin{aligned}\Pi &= TR - TC \\ &= P \times Q - ATC \times Q \\ &= (P - ATC) \times Q\end{aligned}$$



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2.3.1 If Profit Is Negative, Should a Firm Shut Down?

If the firm shuts down, its profits are:

$$\begin{aligned}\Pi_{SD} &= TR - TC \\ &= TR - (FC + VC) \\ &= 0 - (FC + 0) = -FC\end{aligned}$$

If the firm continues operating, its profits are:

$$\Pi_O = TR - TC = TR - FC - VC$$

It should keep operative if $\Pi_0 > \Pi_{SD}$, or

$$\begin{aligned}\Pi_O - \Pi_{SD} &= TR - FC - VC - (-FC) \\ &= TR - VC\end{aligned}$$

Therefore, in the short run, a firm should operate as long as its revenue is greater than or equal to its variable cost. That's because fixed cost needs to be paid whether the plant operates or not, so it doesn't enter into the operate/shut-down decision.

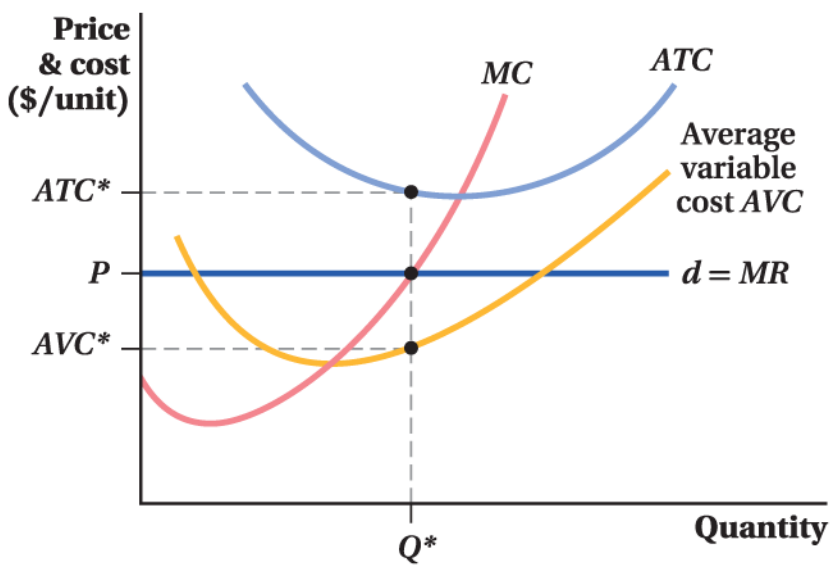

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Figure 2: This firm should keep operating.

3 Perfect Competition in the Short-Run

3.1 A Firm's Short-Run Supply Curve in a Perfectly Competitive Market

Note: the supply curve shows the quantity supplied at any given price, and the firm chooses to produce where $P = MC$, therefore the short-run marginal cost curve is the firm's short-run supply curve. One caveat (shown on the figure below).

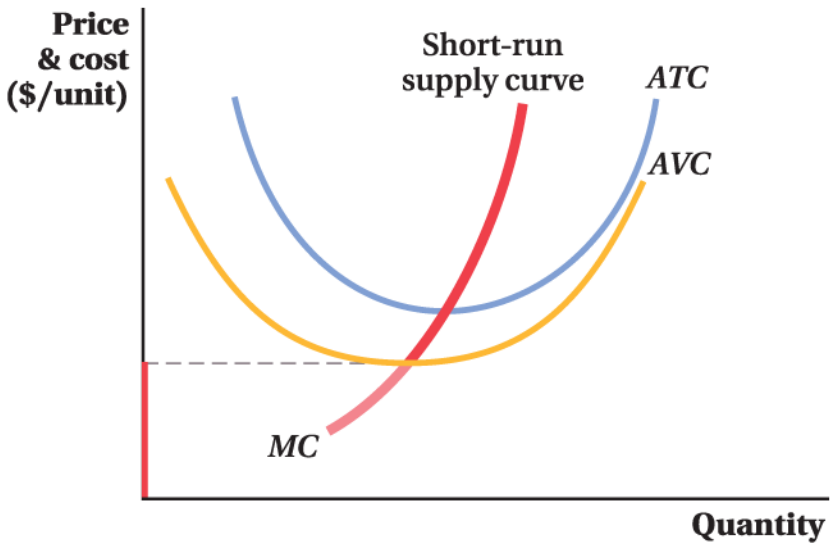

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Figure 3: This firm should keep operating.

POLL EVERYWHERE: The perfectly competitive firm's short-run supply curve is:

3.2 The Short-Run Supply Curve for a Perfectly Competitive Industry

To derive the industry short-run supply curve from these firm supply curves, we add up the individual firm outputs at each possible market price.

3.2.1 The Short-Run Supply Curve: A Graphical Approach

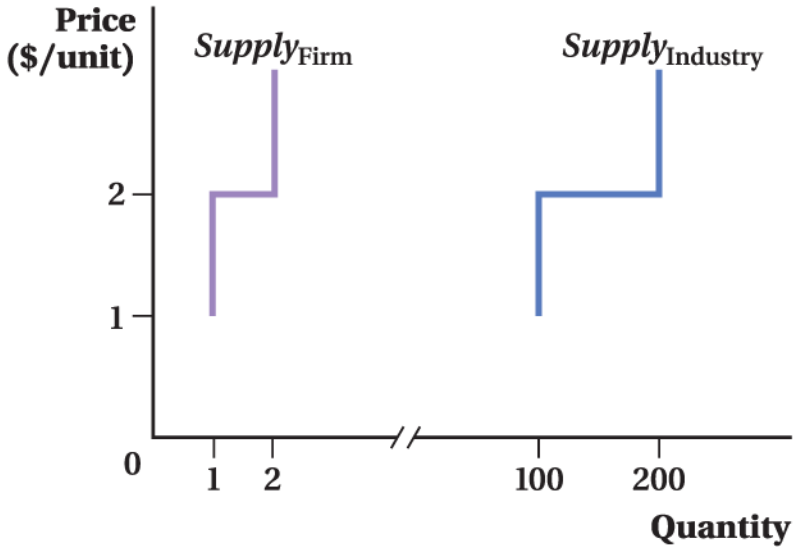

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Figure 4: $Supply_{Firm}$ is the short-run supply curve of each firm in an industry with 100 firms. The short-run industry supply curve, $Supply_{Industry}$, is the horizontal sum of the individual firms' supply curves.

In general, the industry supply curve is the horizontal sum of the supply curves of the individual firms within it.

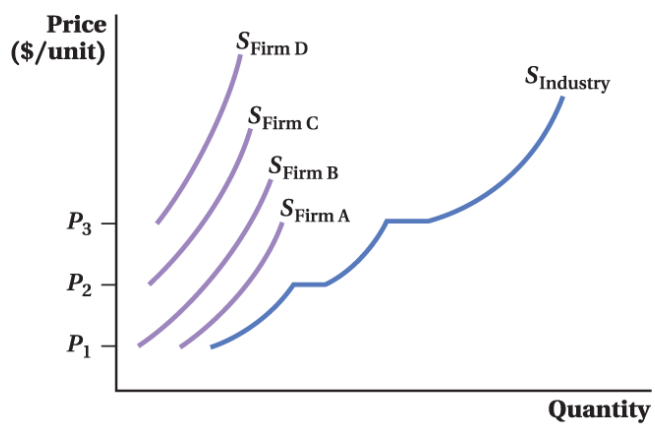

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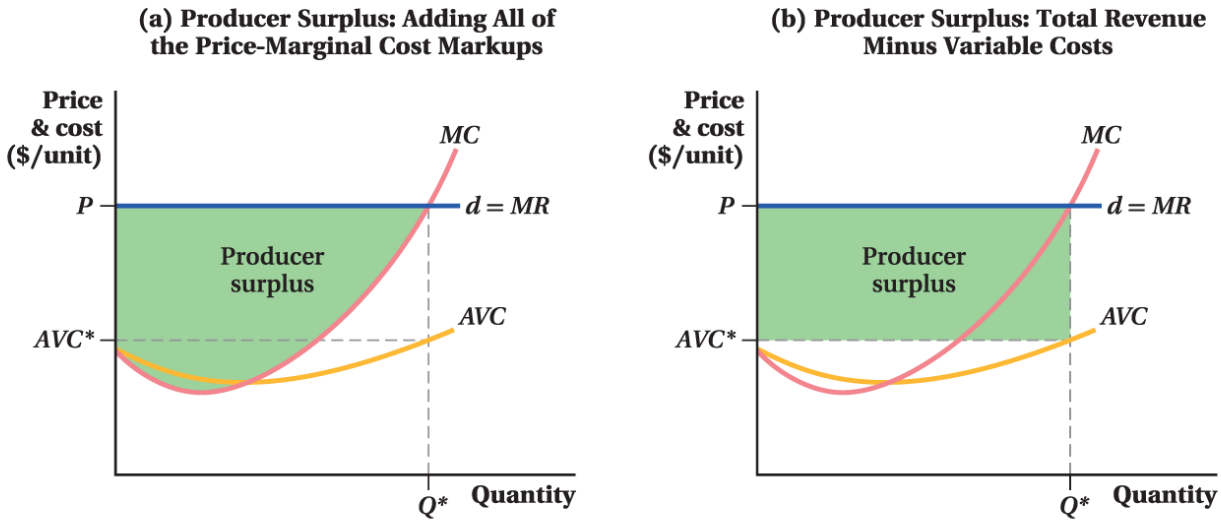
Figure 5: In a four-firm industry in which each firm faces different costs, these cost differences are reflected in their individual supply curves (S_{FirmA}, S_{FirmB}, S_{FirmC}, and S_{FirmD}). The industry supply curve, $S_{Industry}$, is the horizontal sum of the four individual firms' supply curves. At prices between P_1 and P_2, only Firms A and B produce; at price P_2, Firm C also supplies its product on the market; and at price P_3, all four firms supply positive quantities.

3.3 Producer Surplus for a Competitive Firm in the Short-Run

Two ways to compute producer surplus:

- Area between the price and the marginal cost curve.
- Area between the price and the average variable cost at the optimal quantity.
- These two measures are identical.



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3.4 Producer Surplus and Profit

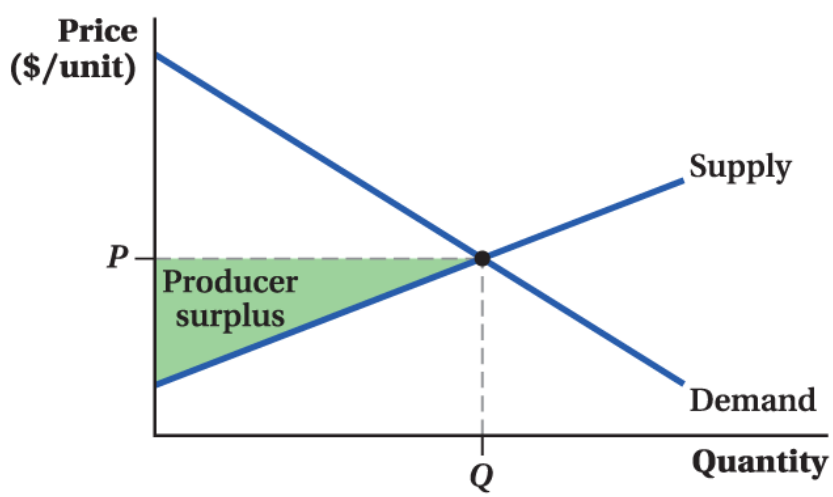
Producer surplus and profit are related but not identical:

$$PS = TR - TVC$$

$$\Pi = TR - TVC - FC$$

A new firm may operate with profit less than zero. It will never operate with producer surplus less than zero because that means each unit costs more to produce than it sells for, even without fixed costs.

3.5 Producer Surplus for a Competitive Industry



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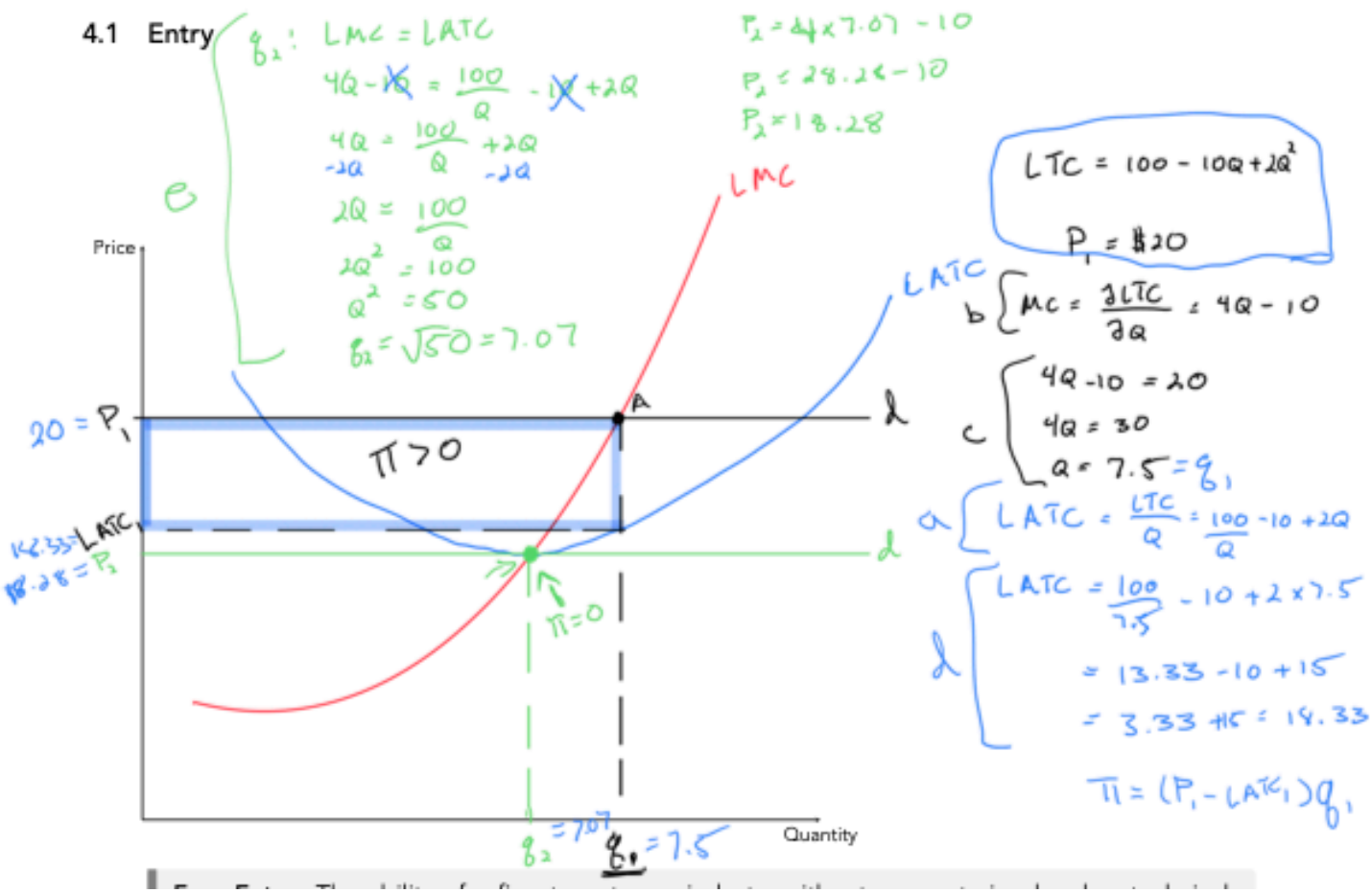
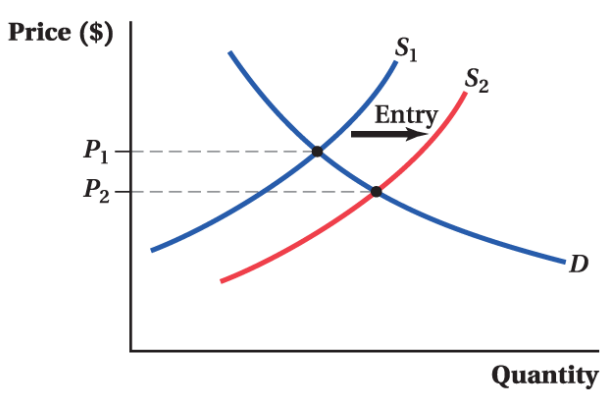
4 Perfectly Competitive Industries in the Long Run

4.1 Entry

> **Free Entry:** The ability of a firm to enter an industry without encountering legal or technical barriers.

Firms decide to enter or exit a market depending on whether they expect their action to be profitable. Think about a firm that is considering entering a perfectly competitive market. For simplicity, assume that all firms in the market, including this and any other potential entrants, have the same cost curves. (We look at what happens when firms have different costs later.)





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> **Long-Run Competitive Equilibrium:** The point at which the market price is equal to the minimum average total cost and firms would gain no profits by entering the industry.

One thing to remember about this no-profit condition is that it refers to economic profit, not accounting profit. The opportunity cost of the business owners' time is included in the firms' costs. It means business owners make only enough to stay in business — that is, to be no worse off than their outside option — and no more.

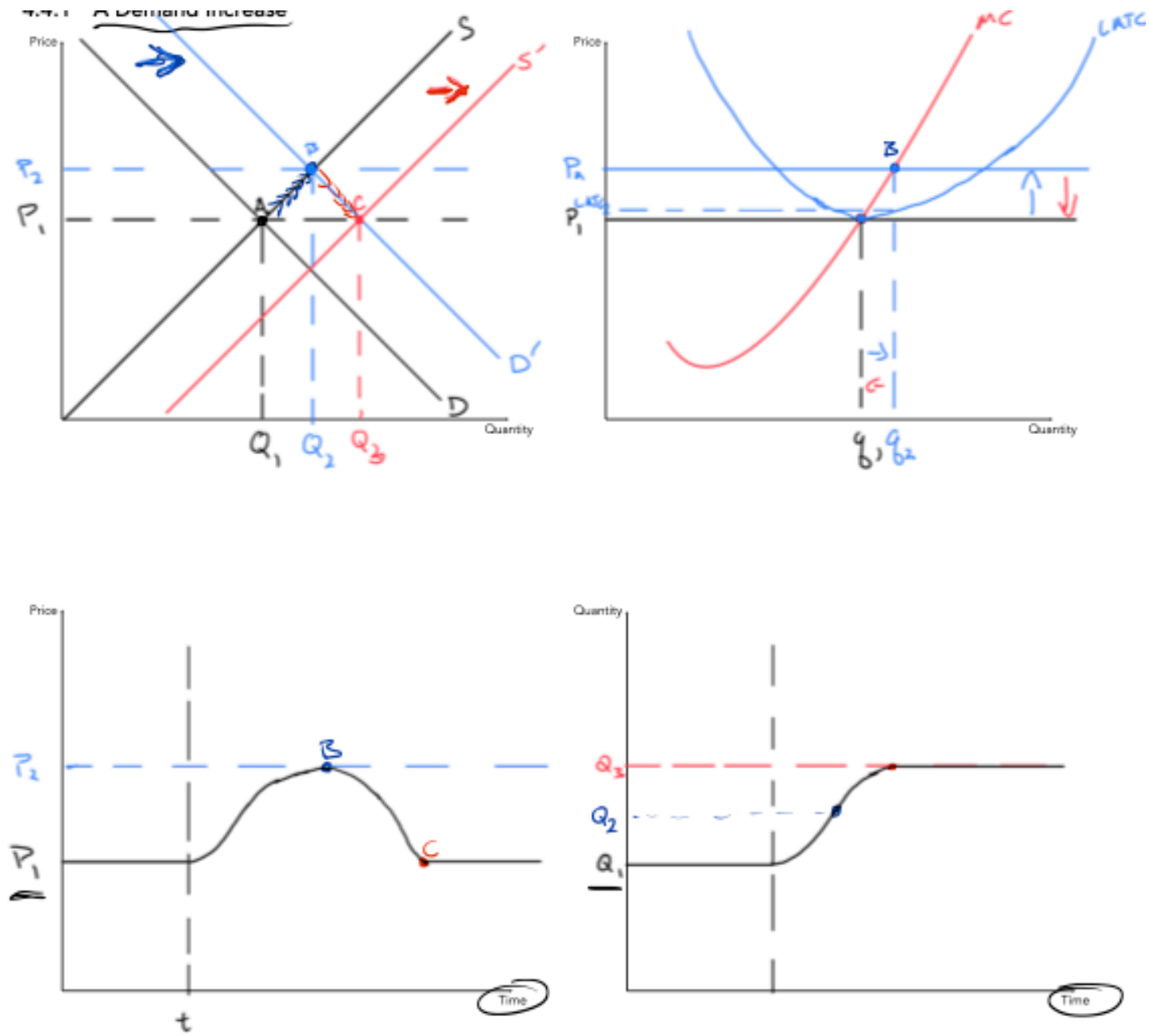
4.2 Exit

> **Free Exit:** The ability of a firm to exit an industry without encountering legal or technical barriers.

This exit from the industry shifts its supply curve in, raising the market price. Exit continues until the market price rises to minimum average total cost. At this point, exiting would not make any firms better off.

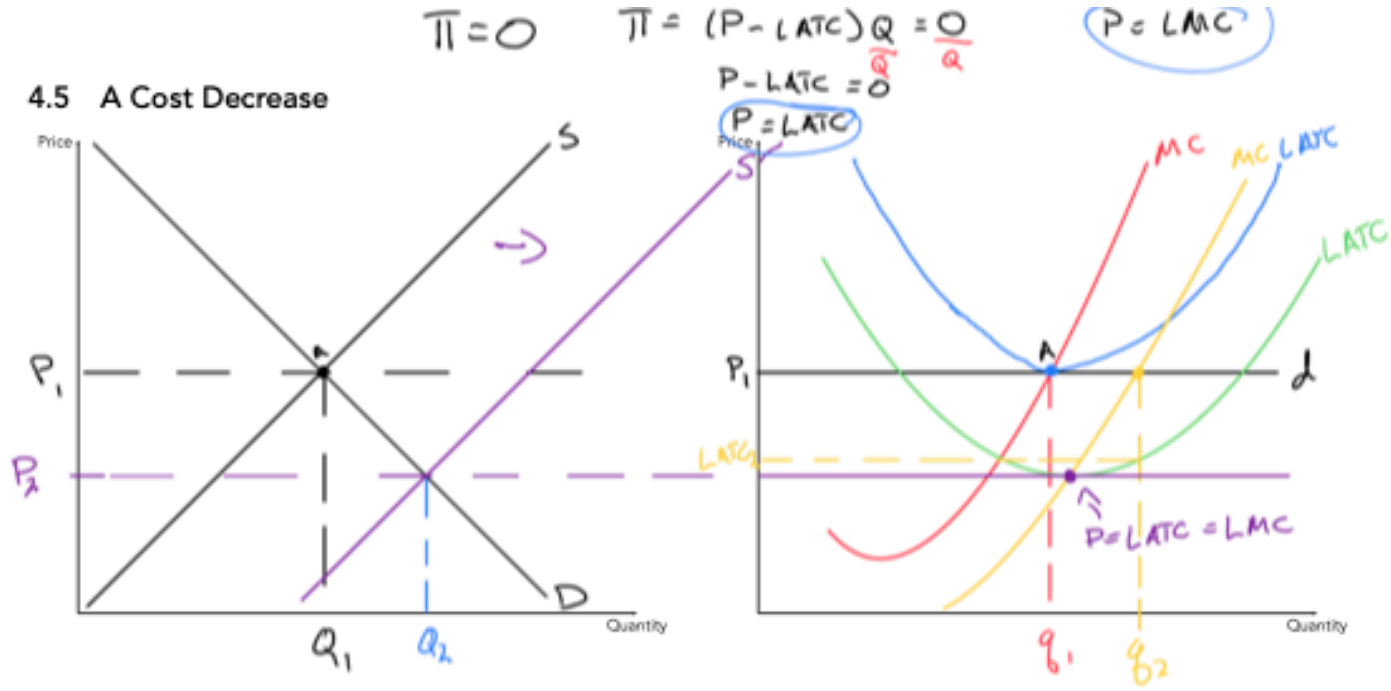
4.3 Adjustments between Long-Run Equilibria

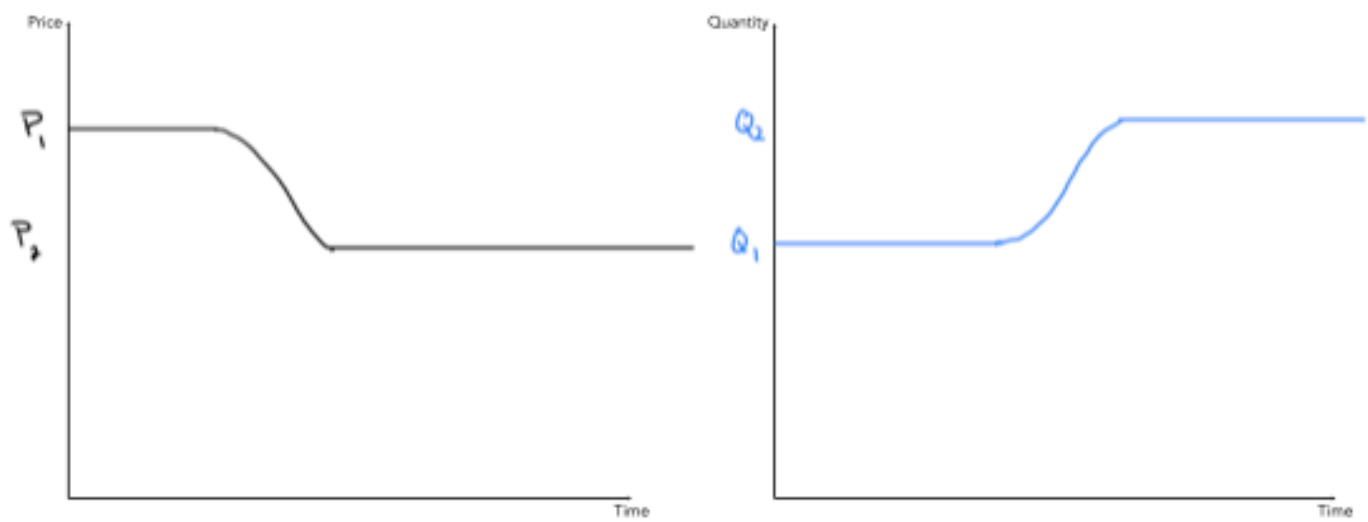
4.3.1 A Demand Increase



- There is an increase in demand which shifts the demand curve to the right.
  - This causes the market price to gradually increase to P_2 and the market quantity to gradually increase to Q_2.
  - The short-run equilibrium is at point B.
- Individual firms respond to this increase in price by increasing their quantity to q_2. The short-run equilibrium is at point B.
  - In the short-run, at point B, the market price is higher than the firm's long-run average cost and therefore the firm is marking profit.
- Since firms are making profit, in the long-run, new firms will enter the market.
  - This will cause the market supply curve to shift to the right and the market price will fall back to P_1.
  - In the long-run, individual firms will produce q_1 units of the good and earn zero economic profits.
  - The market equilibrium quantity will be Q_3 because even though every firm is still producing the same amount of the good as before the increase in demand, there are now more firms.

4.3.2 A Cost Decrease





- There is a cost decrease for the firms which causes the long-run average total cost curve and marginal cost curve to shift down.
  - In the short-run, at the market price P_1, the firms will increase their output to q_2 so that the market price is equal to their new marginal cost (profit maximizing).
  - In the short-run, firms are earning positive economic profits.
- In the long-run, firms will enter the market because of the positive economic profits.
  - This will cause the market supply curve to shift to the right, which will cause the market price to fall.
  - The market price will fall until firms are earning zero economic profits (P_2).

POLL EVERYWHERE: In a perfectly competitive industry, the long-run equilibrium price is $12. If a technological innovation lowers production costs, the long-run equilibrium price will:

4.4 Graphing the Industry Long-Run Supply Curve

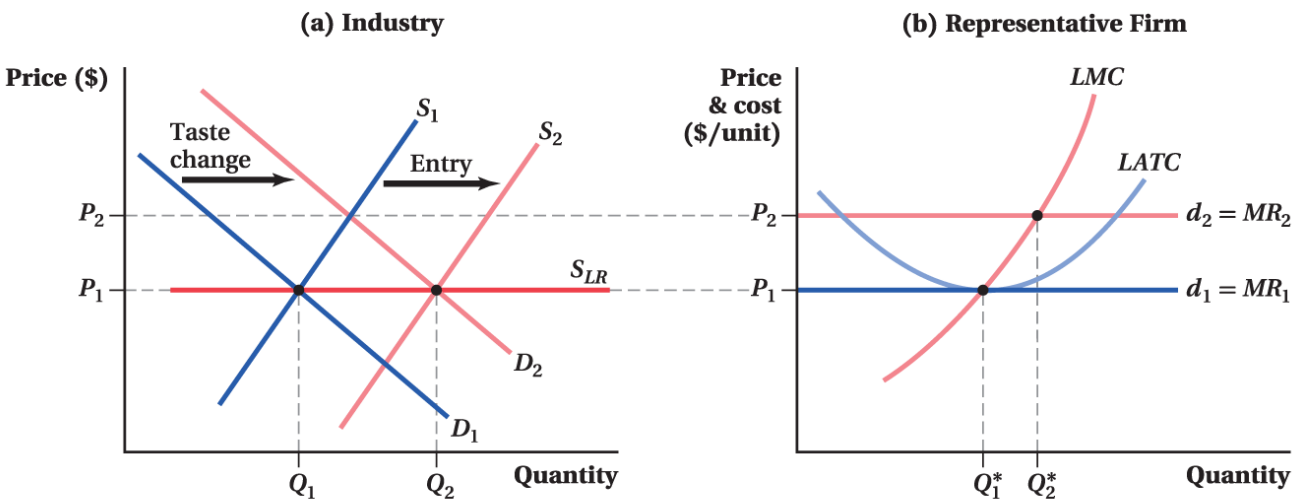


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Figure 6: (a) The original long-run equilibrium for an industry is (P_1, Q_1) at the intersection of the long-run supply curve SLR and the original demand curve D_1. After a change in tastes, demand increases to D_2, price increases to P_2, and firms earn positive economic profits in the short run. In the long run, new firms enter the industry, shifting the short-run supply curve to S_2 until it reaches the long-run equilibrium price P_1 at the new equilibrium quantity Q_2. (b) At the long-run market price P_1, the representative firm earns zero economic profit and produces quantity Q_1^*. When market demand rises, the market price increases to P_2 and the firm's output increases to Q_2^*. At this combination, the firm earns positive economic profit. As entry into the industry occurs, the price falls back to P_1, and the firm reduces its output to Q_1^*. At this point, the firm earns zero economic profit.

4.5 Long-Run Supply in Constant-, Increasing-, and Decreasing-Cost Industries

> **Constant-Cost Industry:** An industry whose firms' total costs do not change with total industry output.

Long-run supply curve is horizontal.

> **Increasing-Cost Industry:** An industry whose firms' total costs increase with increases in industry output.

This might occur because the price of an input rises in response to the industry's higher demand for that input. Suppose an industry requires special capital equipment that is in limited supply. When there is an increase in the industry's output, firms compete for this scarce capital, pushing up its price. This means that the higher industry output, the greater firms' average total costs will be, even in the long run. For this reason, the long-run supply curves of increasing-cost industries are upward-sloping. They're not as steeply sloped as the short-run supply curve for the industry because they account for entry and exit, but they're not horizontal either.

> **Decreasing-Cost Industry:** An industry whose firms' total costs increase with increases in industry output.

This might be the case because there are some increasing returns to scale at the industry level, or in the production of one or more of the industry's inputs. The long-run supply curves for these industries are downward-sloping. Again, the short-run transition between long-run equilibria when demand for the industry's product increases

looks like the constant-industry case, but now entry continues past the point where the market price is driven down to the old long-run average cost level. Instead, entry continues until the price falls all the way to the new, lower minimum average total cost.

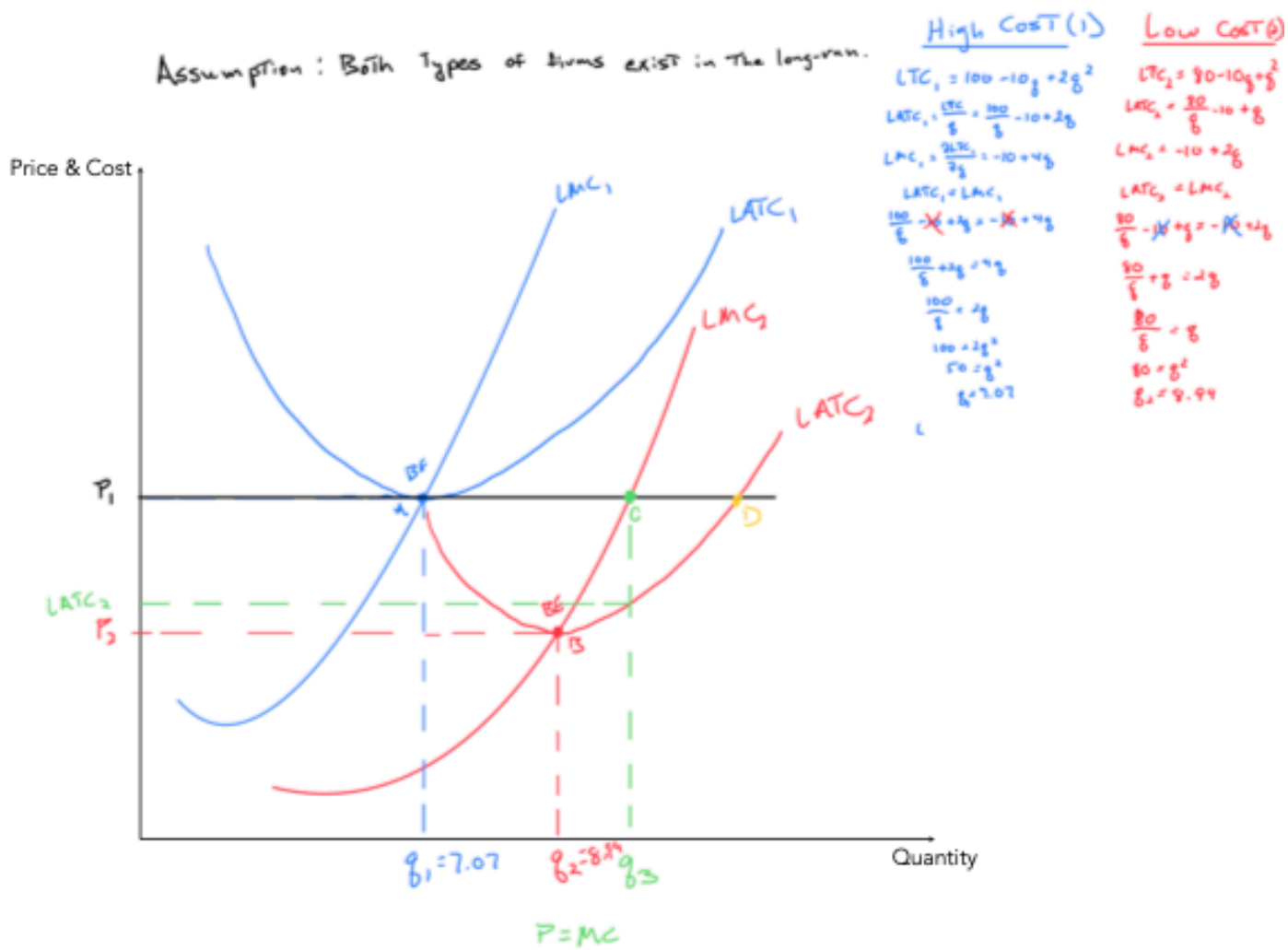
5 Producer Surplus, Economic Rents, and Economic Profits

5.1 Cost Differences and Economic Rent in Perfect Competition

In looking at long-run outcomes in perfectly competitive markets, however, we've assumed that all firms in an industry have the same cost curves. That's not usually true in real life. Firms differ in their production costs for many reasons: They face different prices for their inputs; they have various degrees of special know-how that makes them more efficient; they have a superior location; and so on. When there are cost differences between firms in a perfectly competitive industry, the more efficient producers earn a special type of return called economic rent.

> **Economic Rent:** Returns to specialized inputs above what firms paid for them.

Consider an example in which the factors that cause these firms' costs to differ are specific fixtures of the firm; that is, these costs can't be influenced by the firm's actions or sold to other firms. For instance, the factors might involve access to special technologies no one else has or could use if they were sold. In any case, these factors affect only the cost structure of that particular firm. Figure 8.19 shows the long-run marginal cost curves for three firms.



5.1.1 Economic Profit Is Not the Same As Economic Rent

Firms in perfectly competitive industries make zero economic profit in the long run even if they have different costs. There is a distinction between economic profit and economic rent. Economic profit counts inputs' opportunity costs, and economic rent is included in the opportunity cost for inputs that earn them. This is because inputs that earn rents would still earn them if they were given to another firm. If we gave another firm the brilliant manager or the better location discussed above, it would lower that firm's costs. That other firm would therefore be willing to pay more for that rent-earning input. This willingness to pay for the economic rent inherent to the input raises the opportunity cost of the input to the firm that currently owns it — by using the input, they're giving up the ability to sell it to another firm. Once this opportunity cost is subtracted from the firm's revenue, its economic profit is no higher than if the input earned no rent at all.